

PROCOMP B686 MANUAL

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Chapter 1

Motherboard Feature Introduction

SPECIFICATIONS

System Chipset	Intel® 440LX chip set ,Winbond 83977TF-AW
CPU	Pentium® II Klamath CPU 233MHz ~ 333MHz
Memory Subsystem	Expandable to 384MB(3 banks) with 168-Pin SDRAM(DIMM) Socket X3
Integrated I / O	Two high speed 16550 compatible serial ports, one Multi-Mode Parallel Port fixed SPP/EPP/ECP standard Two PCI Bus master Ultra DMA/33 IDE port (up to 4 IDE Devices) Support two 360KB / 720KB / 1.2MB / 1.44MB / 2.88MB / floppy disk driver Support LS120 drives, ZIP 100 drives Support two USB ports Support IrDA TX / RX header

On-Board W83781D	CPU/Power Supply /chassis Fan Revolution Detect CPU Fan Control (the fan will automatically stop when the system enters suspend mode) CPU Overheat Warning(reserved) Chassis Intrusion Detect (reserved) Display Actual Current Voltage
BIOS	1MB Flash ROM Award AGP BIOS with green, plug and play, ACPI, DMI feature support Support secondary device boot
Expansion slot	Four 32-bit PCI Slots & three 16-bit ISA Slots Support 3.3/5V PCI 2.1 bus Interface
EXTRA Function	Support Keyboard and PS/2 Mouse ON-NOW Function Suspend LED on/off Win95 soft power off External SMI Wake up by ring

Keyboard Connector	PS/2 Keyboard and PS/2 mouse Connector
Others	Windows 95 Compatible
Dimension	ATX size (350mm x 190mm), 4-layer PCB

POWER OFF CONTROL SOFTWARE

The motherboard design supports software power off Control feature through the SMM code in the BIOS under Win95 operating system environment. This is INTEL ATX form factor feature and you should use ATX power supply.

First, you should connect the power switch cable (provided by the ATX case Supplier) to the connector [PW_ON] on the motherboard. In the BIOS screen of “POWER MANAGEMENT SETUP”, choose “User Defined”(or min power saving or Max power saving) in “POWER MANAGEMENT” and choose “Yes” in “PM Control by APM”.

In Windows 95 the “ SHUT DOWN “ option , the computer’s Power will switch off automatically and put the PC in a suspend mode. A blinking power light will indicate this. To restart the system , simply press the Power Button.

PACKAGING CHECK LIST

The motherboard comes securely packed in a durable box and shipping carton. If any of the above items are missing or damaged , please contact your supplier.

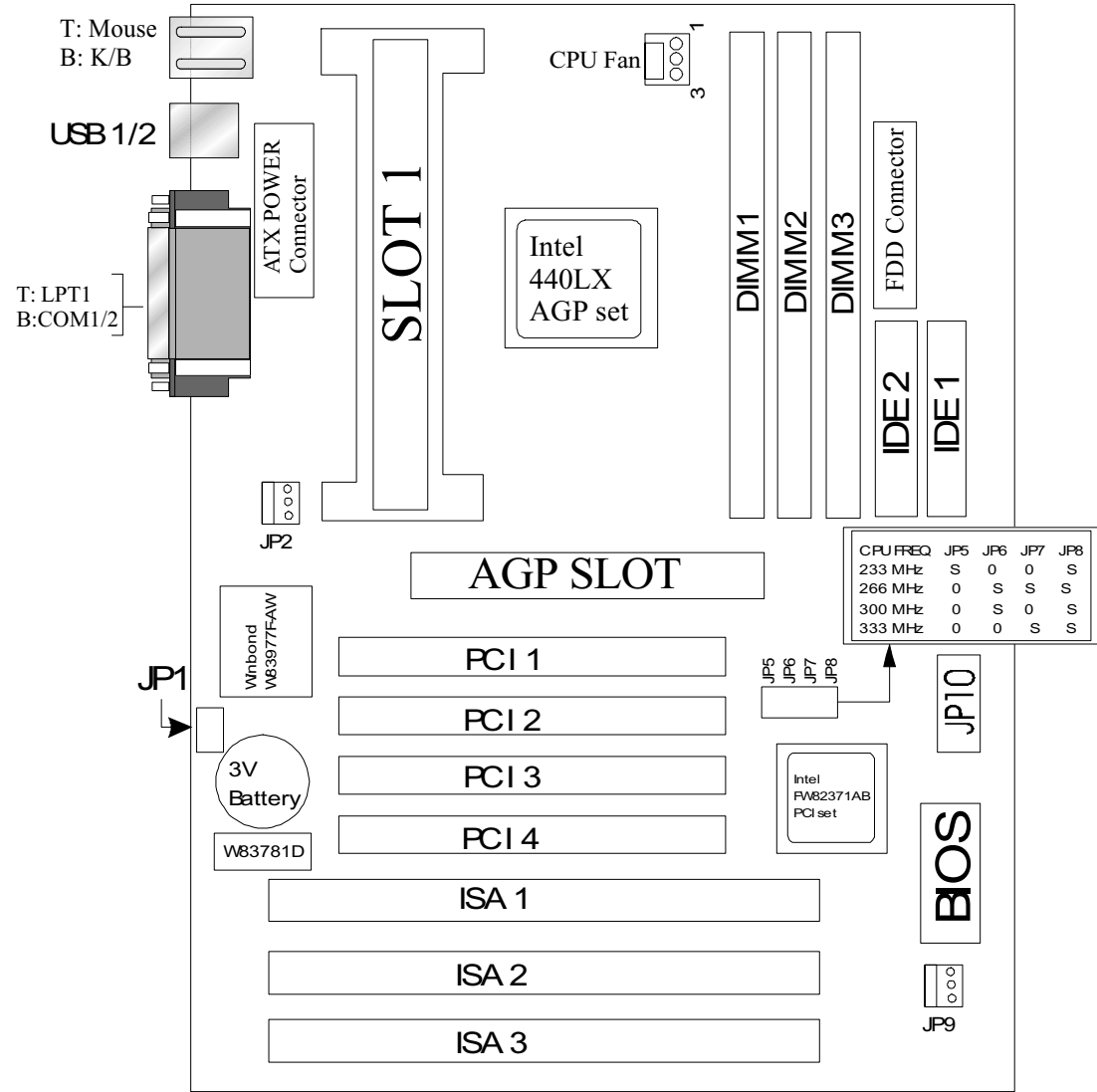
The motherboard contains:

Q' TY	Description
1	Motherboard : B686
1	Diskette : Bus master driver Award system BIOS utility W83781D AP
1	Cable : Enhanced IDE connector
1	Cable : F.D.D connector
1	Manual : User' s manual
1	Temperature Resister : use for temperature sensor

Chapter 2

Setup Guide

Motherboard Layout Drawing



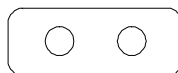
Jumper & Connector Setting

CPU TYPE SELECTION

Processor Core FREQ SYSTEM BUS FREQ		JP5	JP6	JP7	JP8
3.5	233 MHz	Short	Open	Open	Short
4	266 MHz	Open	Short	Short	Short
4.5	300 MHz	Open	Short	Open	Short
5	333 MHz	Open	Open	Short	Short



Short



Open

CONNECTOR SETTING

U1- PS/2 Keyboard Connector

Pin	Description
1	Keyboard Data
2,6	N.C.
3	Ground
4	+5V
5	Keyboard Clock

J4- Power Supply Connector

Pin	Description
1,2,11	+ 3.3 V
3,5,7,13,15,16,17	Ground
4,6,19,20	+ 5 V
8	POWER GOOD
9	5VSB
10	+12 V
12	-12 V
14	PS-ON
18	- 5 V

U2- PS/2 Mouse Connector

Pin	Description
1	Mouse Data
2,6	N.C.
3	Ground
4	+5V
5	Mouse Clock

J1- PRINTER Connector

Pin	Signal Name	Pin	Signal Name
1	Strobe-	14	AFD
2	Data Bit 0	15	Error
3	Data Bit 1	16	INIT
4	Data Bit 2	17	SLCTIN
5	Data Bit 3	18	GND
6	Data Bit 4	19	GND
7	Data Bit 5	20	GND
8	Data Bit 6	21	GND
9	Data Bit 7	22	GND
10	ACK	23	GND
11	Busy	24	GND
12	PE	25	GND
13	SLCT	26	GND

COM1,COM2(PJ1,PJ2) –Serial Connectors

Pin	Signal Name	Pin	Signal Name
1	DCD	6	DSR
2	SIN	7	RTS
3	SOUT	8	CTS
4	DTR	9	RI
5	GND	10	NC

J3(USB1,USB2)- Universal Serial Bus (USB) Connectors

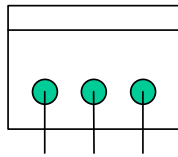
USB1 Pin	Signal Name	USB2 Pin	Signal Name
1	USB VCC 0	1	USB VCC 1
2	USB Data -	2	USB Data -
3	USB Data +	3	USB Data +
4	USB GND 0	4	USB GND 1

J2- Infrared Connector: IR

Pin	Signal Name
1	VCC
2	FIRRX (N)
3	IRRX
4	GND
5	IRTX
7	CIRRX (N)
8	5VSB
6,9,10	NC

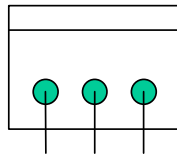
FAN CONNECTOR

JP2



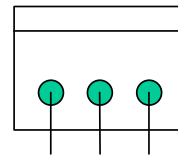
1 2 3
Fan Out +12V FanIN

JP4



1 2 3
GND +12V GND

JP9



1 2 3
GND +12V FanIN

JP2: For CPU COOL FAN CONNECTOR

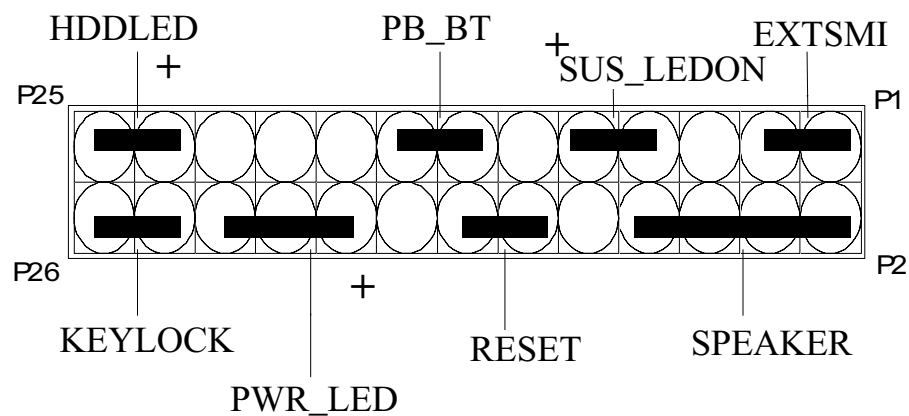
*****this fan can be controlled by w83781D AP ***(on/off)**

JP4: This fan is used in CPU COOL FAN.

JP9: FAN CONNECTOR

*****controlled by w83781D AP*****

JP10 OTHER JUMPER SETTING



Pin	Name	Description
1-3	EXTSMI	Suspend mode
7-9	SUS _ LEDON	Suspend mode LED
13-15	PB_BT	Power buttem
23-25	HDLED	Hard Disk LED
2-8	SPEAKER	Speaker
12-14	RESET	Reset buttom
18-22	PWR _ LED	Power LED
24-26	KEYLOCK	Key Lock

JP1 – CMOS CLEAR

Description	JP4
Normal (default)	1-2
Clear CMOS	2-3

MEMORY INSTALLATION

No jumper setting is necessary for DRAM setting, BIOS will check DRAM type and size automatically. B686 main board contains 3 by 168-pin DIMM sockets (DIMM1,DIMM2,and DIMM3) . B686 main board has table-free (or auto-bank) feature and user can install DIMM into any bank. The three DIMMs Sockets for system memory expansion from 8MB to 384 MB. Each bank provides 64-bit wide data path.

NOTE: Samples of System Memory Combinations Options

DIMM1	DIMM2	DIMM3	TOTAL
8MB	---	---	8Mbytes
---	8MB	---	8Mbytes
---	---	8MB	8Mbytes
8MB	8MB	---	16Mbytes
---	8MB	8MB	16Mbytes
8MB	---	8MB	16Mbytes
16MB	---	---	16Mbytes
---	16MB	---	16Mbytes
---	---	16MB	16Mbytes
8MB	8MB	8MB	24Mbytes
16MB	8MB	---	24Mbytes
16MB	---	16MB	32Mbytes
16MB	16MB	---	32Mbytes
---	---	32MB	32Mbytes
---	32MB	---	32Mbytes
32MB	---	---	32Mbytes
8MB	16MB	16MB	40Mbytes
32MB	32MB	---	64Mbytes
---	32MB	32MB	64Mbytes
64MB	---	---	64Mbytes
64MB	64MB	---	128Mbytes
:	:	:	:
:	:	:	:
128MB	128MB	128MB	384Mbytes

Chapter 3

Award BIOS Setup

Award BIOS ROM has a built-in Setup program that allows users to modify the basic system configuration. This type information is stored in battery-backed RAM so that it retains the Setup information when the power is turned off.

Entering Setup

Power on the computer and press immediately will allow you to enter Setup. The other way to enter Setup is to power on the computer, when the below message appears briefly at the bottom of the screen during the POST (Power On Self Test), press key or simultaneously press <Ctrl>, <Alt>, and <Esc> keys.

TO ENTER SETUP BEFORE BOOT PRESS CTRL-ALT-ESC OR DEL KEY

If the message disappears before you respond and you still wish to enter Setup, restart the system to try again by turning it OFF then ON or pressing the “RESET” button on the system case. You may also restart by simultaneously press <Ctrl>, <Alt> and keys. If you do not press the keys at the correct time and the system does not boot, an error message will be displayed and you will again be asked to,

PRESS F1 TO CONTINUE, CTRL-ALT-ESC OR DEL TO ENTER SETUP

Control Keys

Up Arrow	Move to previous item
Down Arrow	Move to next item
Left Arrow	Move to the item in the left hand
Right Arrow	Move to the item in the right hand
Esc Key	Main Menu Quit and not to save changes to CMOS Status Page setup menu and Option Page Setup Menu Exit current page and return to Main Menu
PgUp Key	Increase the numeric value or make changes
PgDn Key	Decrease the numeric value or make changes
F1 Key	General help, only for Status Page Setup Menu and Option Setup Menu
F2 Key	Change color from total 16 colors
F3 Key	Calendar, only for Status Page Setup Menu
F4 Key	Reserved
F5 Key	Restore the previous CMOS value from BIOS, only for Option Page Setup Menu
F6 Key	Load the default CMOS value from BIOS default table, only for Option Page Setup Menu
F7 Key	Load the default
F8 Key	Reserved
F9 Key	Reserved
F10 Key	Save all the CMOS changes, only for Main Menu

Getting Help

Main Menu

The on-line description of the highlighted setup function is displayed at the bottom of the screen.

Status Page Setup Menu/Option Page Setup Menu

Press F1 to pop up a small help window that describes the appropriate keys to use and the possible selections for the highlighted item. To exit the Help Window press <Esc>.

The Main Menu

Once you enter Award BIOS CMOS Setup Utility, the Main Menu will appear on the Screen.. Use arrow keys to select among the items and press to accept or enter the **sub-menu**.

ROM PC/ISA BIOS (2A69JPNA)
CMOS SETUP UTILITY
AWARD SOFTWARE, INC.

STANDARD CMOS SETUP BIOS FEATURE SETUP CHIPSET FEATURES SETUP POWER MANAGEMENT SETUP PNP/PCI CONFIGURATION LOAD BIOS DEFAULTS LOAD SETUP DEFAULTS	INTEGRATED PERIPHERALS SUPERVISOR PASSWORD USER PASSWORD IDE HDD AUTO DETECTION SAVE & EXIT SETUP EXIT WITHOUT SAVING
Esc : Quit F10 : Save & Exit Setup	←↑↓→ : Select Item (Shift) F2 : Change Color

Standard CMOS Setup

This setup page includes all the items in standard compatible BIOS.

BIOS Features Setup

This setup page includes all the items of Award special enhanced features.

Chipset Features Setup

This setup page includes all the items of chipset special features.

Power Management Setup

This menu provides functions for Green products by allowing users to set the timeout value for monitor and HDD.

PNP / PCI CONFIGURATION SETUP

This menu allows the user to modify PNP / PCI configuration function.

Load BIOS Defaults

BIOS defaults indicates the most appropriate value of the system parameter which the system would be in minimum performance.

Load Setup Defaults

Chipset defaults indicates the values required by the system for the maximum performance.

INTEGRATED PERIPHERALS

This section page includes all the items of IDE hard drive and Programmed Input / Output features.

Supervisor / User Password Setting

Change, set, or disable password. It allows you to limit access to the system and Setup, or just to setup.

IDE HDD Auto Detection

Automatically configure hard disk parameters.

HDD Low Level Format

If supported by your system, this provides a hard disk low level format utility.

Save & Exit Setup

Save CMOS value changes to CMOS and exit setup.

Exit Without Saving

Abandon all CMOS value changes and exit setup.

Standard CMOS Setup

The items in Standard CMOS Setup Menu are divided into several categories. Each category includes no, one or more than one setup items. Use the arrow keys to highlight the item and then use the <PgUp> or <PgDn> keys to select the value you want in each item.

ROM PCI/ISA BIOS (2A69JPNA) STANDARD CMOS SETUP AWARD SOFTWARE, INC.								
Date (mm:dd:yy) : Wed, Dec 28 1994 Time (hh:mm:ss) : 12:35:50								
HARD DISKS	TYPE	SIZE	CYLS	HEAD	PRECOMP	LANDZ	SECTOR	MODE
Primary Master	: Auto	0	0	0	0	0	0	Auto
Primary Slave	: Auto	0	0	0	0	0	0	Auto
Secondary Master	: Auto	0	0	0	0	0	0	Auto
Secondary Slave	: Auto	0	0	0	0	0	0	Auto
Drive A : 1.44M, 3.5 in. Drive B : None 3 Mode : Disable						Base Memory : 640K Extended Memory : 7168K Other Memory : 384K		
Video : EGA/VGA Halt On : All Errors						Total Memory : 8192K		
ESC : Quit				↑↓→← : Select Item		PU/PD/+/-:Modify		
F1 : Help				(Shift) F2 : Change Color				

BIOS Features Setup

ROM PCI/ISA BIOS (2A69JPNA)
BIOS FEATURE SETUP
AWARD SOFTWARE, INC

Virus Warning	: Disabled	Video BIOS Shadow	: Enabled
CPU Internal Cache	: Enabled	C8000-CBFFF Shadow	: Disabled
External Cache	: Enabled	CC000-CFFFF Shadow	: Disabled
Quick Power On Self Test	: Enabled	D0000-D3FFF Shadow	: Disabled
Boot Sequence	: A, C ,SCSI	D4000-D7FFF Shadow	: Disabled
Swap Floppy Drive	: Disabled	D8000-DBFFF Shadow	: Disabled
Boot Up Floppy Seek	: Enabled	DC000-DFFFF Shadow	: Disabled
Boot Up NumLock Status	: On		
Boot Up System Speed	: High		
Gate A20 Option	: Fast		
Typematic Rate Setting	: Disabled		
Typematic Rate (Chars/Sec)	: 6		
Typematic Delay (Msec)	: 250		
Security Option	: Setup		
PCI/VGA Palette Snoop	: Disabled		
OS Select For DRAM > 64MB	: Non-OS2		
		Esc : Quit	↑↓→← : Selection Item
		F1 : Help	PU/PD/+/- : Modify
		F5 : Old Values	(Shift) F2 : Color
		F6 : Load BIOS Default	
		F7 : Load Setup Default	

Virus Warning

This category flashes on the screen. During and after system boots up, any attempt to write to the boot sector or partition table of the hard disk drive will halt the system and the following error message will appear, in the mean time , you can run anti-virus programs to locate the problem.

!WARNING!

Disk boot sector is to be modified

Type “Y” to accept write or “N” to abort write

Award Software, Inc.

Enabled Activate automatically when the system boots up causing a warning message to appear when anything attempts to access the boot sector or hard disk partition table.

Disabled No warning message to appear when anything attempt to access the boot sector or hard disk partition table.

CPU Internal Cache/External Cache

These two categories speed up memory access. However, it depends on CPU/chipset design. The default value is Enabled.

Enabled: Enabled cache

Disabled: Disabled cache

Quick Power On Self Test

This category speeds up Power On Self Test (POST) after you power on the computer. If it is set to Enable, BIOS will shorten or skip some check items during POST.

Enabled: Enable quick POST

Disabled: Normal POST

Boot Sequence

This category determines which drive computer searches first for the hard disk operation system (i.e., DOS).

A, C,SCSI: System will first search for floppy disk drive then second search hard disk driver, then SCSI driver.

C,A,SCSI/ D,A,SCSI/ E,A,SCSI/ F,A,SCSI:

System will first search for IDE hard disk driver (C: D: or E: or F:) then second search floppy disk driver then SCSI hard disk driver.

SCSI,A,C: System will first search SCSI hard disk driver then second search for floppy disk driver then IDE hard disk driver.

CDROM,C,A:

System will first search for the CDROM driver (If the CDROM has a bootable CD title.)and second search hard disk driver then floppy disk driver .

C,CDROM,A:

System will first search for the hard disk driver and second search for CDROM driver (If the CDROM has a bootable CD title,) then searches floppy disk driver.

LS120,C: System will first search LS120 disk driver and second search for IDE hard disk driver.

Swap Floppy Drive

Users can enable this item so that the BIOS will see the hardware “Drive A:” as “Drive B:” , and hardware “Drive B:” as “Drive A:”.

Boot Up Floppy Seek

During POST, BIOS will determine if the Floppy disk drive installed is 40 or 80 tracks. 360 K type is 40 tracks while 720K, 1.2M and 1.44M drive type as they are all 80 tracks.

Enabled: BIOS searches for floppy disk drive to determine if it is 40 or 80 tracks. Note that BIOS can not tell from 720K, 1.2M or 1.44M drive type as they are all 80 tracks.

Disabled: BIOS will not search for the type of floppy disk drive by track number. Note that there will not be any warning message if the drive installed is 360K.

Boot Up NumLock Status

The default value is On.

On: Keypad is number keys

Off: Keypad is arrow keys

Boot Up System Speed

It selects the default system speed – the speed that the system will run at immediately after power up.

High: Set the speed to high

Low: Set the speed to low

Gate A20 Option

The Gate A20 Option default setting is fast.”. This is the optimum setting for this mainboard.

Typematic Rate Setting

This determines the typematic rate.

Enabled: Enable typematic rate

Disabled: Disable typematic rate

Typematic Rate (Chars/Sec)

6 : 6 characters per second

8 : 8 characters per second

10 : 10 characters per second

12 : 12 characters per second

- 15 : 15 characters per second
- 20 : 20 characters per second
- 24 : 24 characters per second
- 30 : 30 characters per second

Typematic Delay (Msec)

When holding the key, the time between the first and second character will be displayed.

- 250 : 250 msec
- 500 : 500 msec
- 750 : 750 msec
- 1000 : 1000 msec

Security Option

This category allows you to limit access to the system and Setup, or just to Setup.

- | | |
|----------------|---|
| System: | The system will not boot and access to Setup will be denied if the correct password is not entered at the prompt. |
| Setup: | The system will boot, but access to Setup will be denied if the correct password is not entered at the prompt. |

Note: *To disable security, select PASSWORD SETTING at Main Menu and then you will be asked to enter password. Do not type anything and just press <Enter>, it will disable security. Once the security is disabled, the system will boot and you can enter Setup freely.*

Video BIOS Shadow

It determines whether video BIOS will be copied to RAM, however, it is optional from chipset design. Video shadow will increase the video speed.

Enabled: Video shadow is enabled

Disabled: Video shadow is disabled

C8000-CBFFF Shadow/DC000-DFFFF Shadow

These categories determine whether optional ROM will be copied to RAM by 16K byte.

Enabled: Optional shadow is enabled

Disabled: Optional shadow is disabled

Chipset Features Setup

ROM PCI/ISA BIOS (2A69JPNA) CHIPSET FEATURES SETUP AWARD SOFTWARE, INC

Auto Configuration	: Enabled	SDRAM CAS latency Time	: 3
DRAM Speed Selection	: 60ns	Auto Detect DIMM/PCI Clk	: Enabled
MA Wait State	: Slow	Spread Spectrum Modulated	: Disabled
EDO RAS# To CAS#	: 3	CPU Warning Temperature	: Disabled
EDO RAS# Precharge Time	: 3	Current System Temp	: 0i aMQP
EDO DRAM Read Burst	: x333	Current CPU1 Temperature	
EDO DRAM Write Burst	: x222	Current CPUFAN1 Speed	X>N>pnk
DRAM Data Integrity Mode	: Non-ECC	Current CPUFAN2 Speed	X>N>pnk
CPU-To-PCI IDE Posting	: Enabled	Current CPUFAN3 Speed	X>N>pnk
System BIOS Cacheable	: Disabled	IN0 (V):	IN1 (V)
Video BIOS Cacheable	: Disabled	IN2 (V):	+ 5 V
Video RAM Cacheable	: Disabled	+12 V	- 12 V
8 Bit I/O Recovery Time	: 1	- 5 V	
16 Bit I/O Recovery Time	: 1		
Memory Hole At 15M-16M	: Disabled	Esc : Quit	↑↓→← : Selection Item
Passive Release	: Enable	F1 : Help	PU/PD/+/- : Modify
Delayed Transaction	: Disabled	F5 : Old Values	(Shift) F2 : Color
AGP Aperture Size (MB)	: 64	F6 : Load BIOS Default	
SDRAM RAS-to-CAS Delay	: Slow	F7 : Load Setup Default	
SDRAM RAS Precharge Time	: Slow		

This setup menu is optimized for this motherboard by your computer vendor. Unless you are a qualified engineer & know the items, functions you are going to modify. We do not recommend you to change the default setting.

Power Management

ROM PCI/ISA BIOS (2A69JPNA)
POWER MANAGEMENT SETUP
AWARD SOFTWARE, INC.

Power Management	: User Define	** Reload Global Timer Events **
PM Control by APM	: Yes	IRQ [3-7,9-15],NMI : Enabled
Video Off Method	: V/H SYNC+Black	Primary IDE 0 : Disabled
Video Off After	: Standby	Primary IDE 1 : Disabled
MODEM Use IRQ	: NA	Secondary IDE 0 : Disabled
Doze Mode	: Disabled	Secondary IDE 1 : Disabled
Suspend Mode	: Disabled	Floppy Disk : Disabled
HDD Power Down	: Disabled	Serial Port : Enabled
Throttle Duty Cycle	: 62.5 %	Parallel Port : Disabled
ZZ Active in Suspend	: Disabled	
VGA Active in Suspend	: Enabled	
Soft-off by PWR-BTTN	: Delay 4 Sec	
CPUFAN off In Suspend	: Enabled	
Resume by Ring	: Disabled	
IRQ 8 Break Suspend	: Disabled	
		ESC: Quit ↑↓→←: Select Item
		F1 : Help PU / PD / + / - : Modify
		F5 : Old Values (Shift)F2 : Color
		F6 : Load BIOS Defaults
		F7 : Load Setup Defaults

This category determines the power consumption for the system after selecting below items. Default value is Disabled. The following pages tell you the options of each item & describe the meanings of each option.

Item	Options	Descriptions
A. Power Management	1. Disable	Global Power Management will be disabled
	2. User Define	Users can configure their own power management
	3. Min Saving	Pre-defined timer values are used such that all timers are in their MAX value
	4. Max Saving	Pre-defined timer values are used such that all timers MIN value

Item	Options	Descriptions
3 PM Control by APM	1. No	System BIOS will ignore APM when power managing the system
	2. Yes	System BIOS will wait for APM' s prompt before it enter any PM mode e.g. DOZE, STANDBY or SUSPEND
	Note: If APM is installed, & if there is a task running, even the timer is time out, the APM will not prompt the BIOS to put the system into any power saving mode!	
	Note: – if APM is not installed, this option has no effect	
	<i>To make the APM function work, users have to install power.exe (supported by MS-DOS 5.0 or higher) in Config.exe. To make the Windows 3.1 work regularly, in “ Windows Setup”, users have to set the “Computer” item to “ MS-DOS System with APM”</i>	
C. Video Off Option	1. Always On	System BIOS will never turn off the screen
	2. Suspend -> Off	Screen off when system is in SUSPEND mode
	3. Susp, Stby -> Off	Screen off when system is in STANDBY or SUSPEND mode

	4. All Modes -> Off	Screen off when system is in DOZE, STANDBY or SUSPEND mode
D. Video	1. Blank Screen	The system BIOS will only blanks off the screen when disabling video
	2. V/H SYN C+Blank	In addition to (1), BIOS will also turn off the V-SYNC & H-SYNC signals form VGA cards to monitor

Item	Options	Descriptions
D. Video	3. DPMS	This function is enabled for only the VGA card supporting DPM
E. HDD Power Down (#) Remark 2	1. Disable	HDD's motor will not off
	2. 1 Min 2 Min 3 Min 4 Min 5 Min 6 Min 7 Min 8 Min 9 Min 10 Min 11 Min 12 Min 13 Min 14 Min 15 Min	Defines the continuous HDD idle time before the HDD entering power saving mode (motor off)
	3. When Suspend	BIOS will turn the HDD's motor off when system is in SUSPEND mode
	Note: - (2) & (3) can't be selected at the same time - When HDD is in power saving mode, any access to the HDD will wake the HDD up	

Item	Options	Descriptions
3 Doze Mode (*) Remark 1	1. Disable	System will never enter DOZE mode
	2. 10 Sec 20 Sec 30 Sec 40 Sec 1 Min 3 Min 5 Min 10 Min 15 Min 20 Min 30 Min 40 Min 1 Hr 2 Hr 3 Hr	Defines the continuous idle time before the system entering DOZE mode. If any item defined in (J) is enabled & active, DOZE timer will be reloaded.
	Note: Normally, STANDBY mode puts the system into low speed or 8 MHz, screen may be off depend on (E)	
3 Standby Mode (*) Remark 1	1. Disable	System will never enter STANDBY mode
	2. 10 Sec 20 Sec 30 Sec 40 Sec 1 Min 3 Min 5 Min 10 Min 15 Min 20 Min 30 Min 40 Min 1 Hr 2 Hr 3 Hr	Defines the continuous idle time before the system entering STANDBY mode. If any item defined in (J) is enabled & active, STANDBY timer will be reloaded
	Normally, STANDBY mode puts the system into low speed or 8, screen may be off depend on (E)	

Item	Options	Descriptions
H. Suspend Mode (*) Remark 1	1. Disable	System will never enter SUSPEND mode
	2. 10 Sec 20 Sec 30 Sec 40 Sec 1 Min 3 Min 5 Min 10 Min 15 Min 20 Min 30 Min 40 Min 1 Hr 2 Hr 3 Hr	Defines the continuous idle time before the system entering SUSPEND mode. if any item defined in (J) is enabled & active, SUSPEND timer will be reloaded
	Note: Normally, SUSPEND mode puts the system into low speed or 8 MHz, clock is stopped, screen may be off depend on (E)	
I. PCI Master Activity COM Ports Activity LPT Ports Activity HDD Ports Activity DMA Ports Activity VGA Activity IRQ3 (COM 2) IRQ4 (COM 1) IRQ5 (LPT 2) IRQ6 (Floppy Disk) IRQ7 (LPT 1) IRQ8 (RTC Alarm) IRQ9 (IRQ2 Redir) IRQ10 (Reserved) IRQ11 (Reserved) IRQ12 (PS/2 Mouse) IRQ13 (Coprocessor) IRQ14 (Hard Disk) IRQ15 (Reserved)	1. Disable	The specified event's activity will not affect the PM timers
	2. Enable	The specified event's activity causes the PM Timers to be reloaded. i.e. the Power Management Unit(PMU) monitors the specified activities as PM events

* Remark 1: All items mark with (*) in this menu, will be loaded with predefined values as long as the item "Power Management" is not configured to "User Defined"

These items are:

Item "System Doze", "System Standby" & "System Suspend"

Remark 2: Although the item "HDD Power Down" is not controlled by item "Power Management" in terms of timer value, the HDD (s) will not power down if the global power management is disabled!

PNP / PCI Configuration Setup

ROM PCI/ISA BIOS(2A69JPNA)
PNP/PCI CONFIGURATION
AWARD SOFTWARE, INC.

PNP OS Installed : No	PCI IDE IRQ Map TO :ISA
Resources Controlled By : Auto	
Reset Configuration Data : Disabled	
	ESC: Quit ↑↓→←: Select Item F1 : Help PU / PD / + / - : Modify F5 : Old Values (Shift)F2 : Color F6 : Load BIOS Defaults F7 : Load Setup Defaults

The following pages tell you the options of each item & describe the meanings of each option.

Item	Options	Descriptions
A. 1st Available IRQ	3	The system BIOS will assign these 4 available IRQs to the found PCI devices
2nd Available IRQ	4	
3rd Available IRQ	5	
4th Available IRQ	7	
	9	
	10	
	11	
	12	
	14	
	15	
	NA	

Item	Options	Descriptions
B. PCI IDE 2nd Channel	Enable Disable	Enable/disable 2nd channel of PCI/IDE card. It includes I/O port (170H~177H) and IRQ 15 assignment
C. PCI IDE IRQ Map To	PCI-AUTO PCI-SLOT1 PCI-SLOT2 ISA	<u>PCI-AUTO</u> The BIOS will: – scan for PCI IDE devices & determine the location of the PCI IDE device
	PCI-AUTO PCI-SLOT1 PCI-SLOT2 ISA	<u>PCI-SLOT1</u> <u>PCI-SLOT2</u> – assign IRQ 14 for primary IDE INT# IRQ 15 for secondary IDE INT# for the specified slot <u>ISA</u> – The BIOS will not assign any IRQs even if PCI IDE card is found! Because some IDE cards connect the IRQ 14 & 15 directly from ISA slot through a cord. (This cord is called Legacy Header)
F. Primary IDE INT# Secondary IDE INT#	A B	To tell which INT# does the PCI IDE card is using for its interrupts

Your computer vendor optimizes the other items, please do not modify them unless you know its function exactly.

INTEGRATED PERIPHERALS

ROM PC/ISA BIOS(2A69JPNA)
INTEGRATED PERIPHERALS
AWARD SOFTWARE, INC.

IDE HDD Block Mode	: Enable	UART Mode Select	: Normal
IDE Primary Master PIO	: AUTO		
IDE Primary Slave PIO	: AUTO		
IDE Secondary Master PIO	: AUTO	Onboard Parallel Port	: 378H/IRQ
IDE Secondary Slave PIO	: AUTO	Parallel Port Mode	: SPP
IDE Primary Master UDMA	: AUTO		
IDE Primary Slave UDMA	: AUTO		
IDE Secondary Master UDMA	: AUTO		
IDE Secondary Slave UDMA	: AUTO		
On-Chip Primary PCI IDE	: Enabled		
On-Chip Secondary PCI IDE	: Enabled		
USB keyboard Support	: Disabled		
POWER ON Function	: Button Only		
KBC input clock	: 8MHZ		
Onboard FDC Controller	: Enabled		
Onboard Serial 1	: 3F8/IRQ4	Esc : Quit	↑↓→← : Selection Item
Onboard Serial 2	: 2F8/IRQ3	F1 : Help	PU/PD/+/- : Modify
		F5 : Old Values	(Shift) F2 : Color
		F6 : Load BIOS Default	
		F7 : Load Setup Default	

This setup menu is optimized for this motherboard by your computer vendor. Unless you are a qualified engineer & know the items, function you are going to modify. We do not recommend you to change the default setting.

|| This Motherboard (B686) can support “ ON-NOW” Function, so the item of “POWER ON Function” is Enabled.

Load BIOS Default

When you access "Load BIOS Default", the following message appears:

Load BIOS Default (Y/N) ?N

The BIOS Default values are the "worst case" default, and are the most stable values for the system. Use them if the system is performing erratically due to hardware problems. To load the BIOS Default values, press <Y> then <Enter>.

Load Setup Default

When you access "Load Setup Default", you are shown the following message:

Load Setup Default (Y/N) ?N

The Setup Default values represent the "best case" defaults, and should provide optimum system performance. To load the Setup Default values, press <Y> then <Enter>.

Supervisor / User Password Setting

When you select this function, the following message will appear at the center of the screen to assist you in creating a password.

ENTER PASSWORD

Type the password, up to eight characters, and press <Enter>. The password typed now will clear any previously entered password from CMOS memory. You will be asked to confirm the password. Type the password again and press <Enter>. You may also press <Esc> to abort the selection and not enter a password.

If you select System at Security Option of BIOS Features Setup Menu, you will be prompted for the password every time the system is rebooted or anytime you try to enter Setup. If you select Setup at Security Option of

BIOS Features Setup Menu, you will be prompted only when you try to enter Setup.

IDE HDD Auto Detection

This feature allows you to check all the information on your hard disk formation. When you access "IDE HDD Auto Detection", the system executes auto detection.

At the prompt, it represents all the information on your HDD, and you are asked:

Do you accept this drive C: (Y/N) ?

- 1 If you accept the test result, press [Y] then [Enter] and the result is saved, then the system continues to detect another HDD.
- 2 If not, press [N] then [enter] and the system continues to detect another HDD.

Winbond W83781D Setup Guide

The w83781D supports 3 Temperature Resister Sensors , voltage detection, and fan speed sensor control.

1. 3 Temperature Resister Sensors:

- a. RT2: This function is the CPU's temperature sensor.
- b. RT1/RT3: These functions are the temperature sensor for the surrounded PC environment.
- c. You can connect the temperature resister wire to the RT2 socket in order to sensor the CPU's temperature, when you have placed the temperature resister wire , tape the wire on top of the CPU heat sink.
- d. You can connect the temperature resister wire to the RT1 socket in order to control the system fan (JP9),when over temperature 60 degree (default),the fan will turn on.
- e. If you put the resister on top of RT1,RT2,RT3, you will be able to read the temperature on W83781D AP.

2. Fan speed sensor control:

- a. JP2 is the CPU's Fan speed sensor control.
- b. JP9 is the system Fan.
- c. JP2 and JP9 is an optional function.

3. Voltage detection:

You can see the Voltage detection on W83781D AP.

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